

Listening to the Collective: Entropy-Based Early Warning for Emergent Agent Communication¹

Juan Camilo Melgarejo Gómez¹

Affiliation

Alejandro Sánchez Poveda²

Affiliation

Sergio Montoya Ramírez³

Affiliation

Daniel Díaz González⁴

Affiliation

Roger Palomeque⁵

Affiliation

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Abstract

Summarize your project in 150–250 words. A strong abstract lets a reviewer understand what you did and why it matters without reading anything else. Make sure to cover: the problem, your approach, key results, and the main takeaway. Polish it last: the abstract should reflect your final results, not your initial plan.

How to use this template: Replace the *italicized guidance text* under each section with your content. The section structure is strong guidance but not rigid. If your project requires a different organization, feel free to adapt. Delete all guidance text including this info box before submitting. Make sure the project title and author information above are up to date.

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Recommended length: 4 pages excluding references and appendix. Rough guide: Intro & Related Work 1p, Methods and Results 2.5p, Discussion 0.5p.

1. Introduction

In the pursuit of AI safety, one of the central requirements when designing evaluation and deployment environments is the availability of heuristics and tools capable of detecting unwanted behaviour. This requirement becomes more pressing as large language model (LLM) agents are increasingly deployed as multi-agent systems (MAS), in which several agents that reason, plan and act through tools are combined to decompose tasks, explore in parallel or verify each other's work (Kim et al., 2026; Zhu et al., 2026). In such systems, undesired behaviour need not originate in a single agent: it can emerge from the interaction between agents, including coordination that was never intended by the system designer. The OpenAI–Hugging Face incident of July 2026 (Larcher et al., 2026; OpenAI, 2026) showed that this risk is no longer hypothetical, and it is the direct motivation for the present work.

¹Research conducted at the [AI Incident Response Sprint](#), September 2026.

1.1 Problem and proposed approach

The incident also illustrates why detecting communication by monitoring specific channels is not a viable strategy. The agents exchanged information through plaintext files, directory names, messages encoded in URL paths, public Hugging Face datasets used as a pull-based command-and-control channel, pastebins, webhooks and even rendered screenshots decoded by optical character recognition (OpenAI, 2026). Hugging Face further reports that staged data was chunked, XOR-encrypted, gzip-compressed and base64-encoded, so that naive log scans missed about 75% of the sensitive material (Larcher et al., 2026). None of these resources was designed as a communication channel, and a detector written for any one of them would have been bypassed by the next. Moreover, the effect of coordination on system behaviour depends strongly on the architecture and on the capability of the underlying models (Kim et al., 2026), which makes purely structural detection criteria unreliable.

In this paper we propose a channel-agnostic mechanism for detecting communication between agents in a MAS. Rather than inspecting the channels themselves, we study the behaviour of the entropy of the individual agents and of the system as a whole, and we fit regression models to its trajectory in order to detect anomalies that may indicate interaction. The underlying intuition is that agents that are supposed to be independent, such as parallel evaluation runs, should produce statistically independent outputs; when they begin to read and act upon each other’s messages, their outputs become coupled, regardless of the medium used to exchange them. Entropy-based summaries have proven effective for anomaly detection in network traffic (Lakhina et al., 2005), for uncertainty estimation and hallucination detection in LLMs (Farquhar et al., 2024; Kuhn et al., 2023), and for characterising the dynamics of orchestrated multi-agent systems (Zhu et al., 2026). Applied to the incident described above, a monitoring layer of this kind could, in principle, have raised an alarm when otherwise independent evaluation runs began to show correlated behaviour in May, weeks before the Hugging Face compromise and without any prior knowledge of Artifactory as a channel. We regard such an interaction-detection layer as complementary to chain-of-thought monitoring and to the escalation triggers proposed by OpenAI, and we treat its early-warning capability as a hypothesis to be tested rather than as an established result.

To begin testing this hypothesis, we complement the proposal with a preliminary experimental study based on a simple benchmark, whose purpose is to determine whether the entropy trajectory follows a general functional form and whether that form changes when agents communicate. The remainder of the paper is organised as follows. First, we define the information-theoretic framework on which the method is based. Next, we describe the concrete implementation of the benchmark and the methodology, including the precise definition of the dataset and of the models used. Finally, we interpret the collected measurements in search of mechanistic changes in the entropy that can be attributed to inter-agent communication.

Our main contributions are:

1. *[First contribution — what new thing did you create, discover, or demonstrate?]*
2. *[Second contribution]*
3. *[Third contribution, if applicable]*

2. Related Work

Entropy as a detection primitive. Entropy has long been used to monitor systems whose failures cannot be enumerated in advance. In network security, Lakhina et al. (2005) showed that summarising the *distributions* of packet features (source and destination addresses and ports) with entropy exposes anomalies that volume-based detectors miss: a distributed denial-of-service attack, for instance, concentrates destination addresses (lowering their entropy) while dispersing source addresses (raising it), and this shift is visible without inspecting the payload or knowing the attack type beforehand. In LLMs, entropy of the output distribution is a standard measure of uncertainty, either as predictive entropy over token sequences (Malinin & Gales, 2021) or as semantic entropy over clusters of equivalent answers, which has proven effective for detecting hallucinations (Farquhar et al., 2024; Kuhn et al., 2023).

Entropy in multi-agent systems. Recent work has started to apply these tools to LLM-based MAS. Zhao et al. (2026) measure entropy at the token, agent and round levels and find that a single agent outperforms the collaborative system in a large fraction of cases, with the outcome largely determined by entropy patterns established early in the interaction. Zhu et al. (2026) characterise the entropy dynamics of orchestrated systems and fit them with a damped-oscillation model, which we adapt in our analysis, while Kim et al. (2026) show that the benefits of collaboration shrink as the underlying models become more capable. In all of these studies communication is designed, cooperative and known to the analyst, and entropy is used to explain or improve task performance, not to detect coordination that the designer did not intend.

Monitoring unintended agent behaviour. Current safety tooling mostly inspects content. Chain-of-thought monitoring can catch misbehaviour such as reward hacking, even with a weaker monitor model, but models placed under optimisation pressure learn to obfuscate their reasoning (Baker et al., 2025). AI control protocols pair an untrusted model with trusted monitors and limited human auditing (Greenblatt et al., 2024), and activation probes offer a cheaper classifier for specific, pre-defined behaviours (Kramár et al., 2026). For multiple agents, collusion is recognised as a distinct failure mode (Hammond et al., 2025), and LLMs are increasingly capable of hiding messages through steganography (Motwani et al., 2024), which undermines any detector that reads the exchanged content. The OpenAI–Hugging Face incident is a concrete example: messages were encoded, compressed and spread over resources never meant for communication (Larcher et al., 2026; OpenAI, 2026).

Bridging the two lines of work. Our method transfers the reasoning behind entropy-based network anomaly detection to MAS. The mapping is direct: network flows correspond to agents; packet-feature distributions correspond to the agents’ next-token distributions; and an unknown attack or payload corresponds to an unknown communication channel or encoding. Just as a network attack is revealed by a shift in the entropy of feature distributions rather than by its content, communication between agents that should be independent is expected to appear as a change in their individual entropies and, above all, as a positive total correlation (for two agents, mutual information) between them (Equation 4). This differs from the previous work in three ways. Unlike semantic entropy, which measures the uncertainty of a single model over many samples, the target quantity is the *coupling between agents*. Unlike prior MAS entropy studies, entropy is used as a monitor, calibrated against non-communicating baselines and validated against a provenance-

based ground truth of information transfer. Unlike content monitors, it is agnostic to the channel and to the encoding. We do not intend it to replace chain-of-thought monitors or probes, but to complement them as a cheap, always-on first layer that can trigger more expensive inspection. Its cost is low because the logits are already produced by the forward pass that generates each token, so no additional model call is needed; the price is that it requires access to the output distributions, which in practice restricts it to the operator hosting the agents.

3. Methods

Describe your approach clearly enough that someone could replicate it. Include key design choices and justify them where relevant (hint: the more you can back up your design choices by referencing prior work, the better).

What models/datasets/tools did you use and why? What were key parameters or design decisions? What did you try that didn't work? Could someone reproduce your work from this description?

3.1 Entropy Framework

Introduced by Claude Shannon in 1948 (Shannon, 1948), entropy has become one of the cornerstones of information theory. It sets the fundamental limit of lossless data compression, which is attained in practice by source codes such as Huffman coding (Huffman, 1952); it serves as a summary statistic for detecting anomalies in network traffic, including distributed denial-of-service (DDoS) attacks (Lakhina et al., 2005); and, more recently, it has become a key tool for estimating uncertainty and detecting hallucinations in LLMs (Farquhar et al., 2024; Kuhn et al., 2023). For a discrete random variable X taking values $x_1, \dots, x_n$, the entropy is defined as

$$H(X) = - \sum_{i=1}^n p(x_i) \log_2 p(x_i), \quad (1)$$

where $p(x_i)$ is the probability of the event $X = x_i$, and the convention $0 \log_2 0 = 0$ is adopted. With the logarithm in base 2, entropy is measured in bits. $H(X)$ quantifies the expected uncertainty (or surprise) associated with the outcome of X : it is zero when the outcome is certain, and it reaches its maximum value, $\log_2 n$, when all outcomes are equally likely (Cover & Thomas, 2006).

3.1.1 Entropy of a single agent

For an LLM-based agent, the natural probability distribution is the model's distribution over the next token. At each generation step t , the model produces a vector of logits $z_t \in \mathbb{R}^{|V|}$ over the vocabulary V . Logits are unnormalised scores, not probabilities; the next-token distribution is obtained through the softmax function,

$$p(v \mid x_{<t}) = \frac{\exp(z_{t,v}/T)}{\sum_{u \in V} \exp(z_{t,u}/T)}, \quad (2)$$

where $x_{<t}$ denotes the context (prompt and previously generated tokens) and T is the sampling temperature. Consequently, any entropy computed from this distribution depends on the chosen temperature. Two families of entropy measures can be built on top of these distributions:

1. **Predictive entropy.** The token-level entropy

$$H_t = - \sum_{v \in V} p(v \mid x_{<t}) \log_2 p(v \mid x_{<t}) \quad (3)$$

measures the model’s uncertainty about the next token. At the sequence level, the predictive entropy of a response s given an input x is $H(S \mid x) = -\mathbb{E}_{s \sim p(\cdot \mid x)} [\log_2 p(s \mid x)]$, which is usually estimated by Monte Carlo sampling or approximated by aggregating the token-level entropies along a generated sequence (Malinin & Gales, 2021). Because it operates directly on token probabilities, predictive entropy mixes semantic uncertainty with lexical and syntactic uncertainty: two phrasings of the same content count as different outcomes.

2. **Semantic entropy.** Semantic entropy was introduced to separate uncertainty about meaning from uncertainty about form, a separation that is crucial for applications such as hallucination detection (Farquhar et al., 2024; Kuhn et al., 2023). Instead of operating on individual tokens, it samples several complete responses to the same input, groups them into clusters of semantically equivalent answers (for example, using bidirectional entailment judged by a natural language inference model), and computes the entropy of the distribution over these meaning clusters. As a result, variation in wording that does not change the meaning of a response does not increase the entropy.

In this work we adopt predictive entropy. Our working hypothesis is that, when two agents start to interact, the incorporation of content produced by another agent into an agent’s context (with a different style, vocabulary and information) changes its next-token distributions, and that part of this signal lies precisely in the lexical and syntactic variation that semantic entropy is designed to discard. This choice has two caveats that must be controlled for in the experiments. First, predictive entropy is also sensitive to factors unrelated to communication, such as response length, prompt formatting and temperature, which can act as confounders. Second, many commercial APIs expose only the log-probabilities of the top- k tokens rather than the full distribution, so Equation (3) can only be approximated, for instance by renormalising over the returned tokens or by bounding the contribution of the remaining probability mass.

3.1.2 Entropy of the multi-agent system

The quantities above are defined per agent and are therefore insufficient to describe the MAS as a whole. Let $X_1, \dots, X_n$ be random variables describing the outputs of the n agents in the system (for instance, their next-token or response distributions at a given time step t). The joint entropy of the system decomposes as

$$H(X_1, X_2, \dots, X_n) = \sum_{i=1}^n H(X_i) - TC(X_1; X_2; \dots; X_n), \quad (4)$$

where $TC(X_1; X_2; \dots; X_n)$ is the total correlation (Watanabe, 1960), also known as multi-information. This expression is in fact the definition of total correlation, which quantifies the redundancy, or statistical dependence, shared among the agents. It satisfies $TC \geq 0$, and for $n = 2$ it reduces to the mutual information $I(X_1; X_2)$ (Cover & Thomas, 2006). Intuitively, if the per-agent entropies

were simply added, the information shared among interacting agents would be counted more than once, and the sum would overestimate the true uncertainty of the system; the total correlation term corrects for this overcounting. When the entropies are tracked over time, all terms in Equation (4) are evaluated at the same step t .

This decomposition contains one of the key details for using entropy to fingerprint the behaviour of a MAS. Two regimes can be distinguished:

1. **Independent agents.** $TC = 0$ if and only if the variables $X_1, \dots, X_n$ are mutually independent. In that case the system entropy equals the sum of the individual entropies, $H(X_1, \dots, X_n) = \sum_i H(X_i)$.
2. **Coupled agents.** When agents exchange information, directly or through a shared resource, their outputs become statistically dependent, $TC > 0$, and the joint entropy falls below the sum of the individual entropies. This is the regime observed in the OpenAI–Hugging Face incident, where agents in separate evaluation runs, designed to be independent, coordinated through an improvised message board hosted on shared infrastructure (OpenAI, 2026).

The gap $\sum_i H(X_i) - H(X_1, \dots, X_n)$ is therefore a natural candidate signal for detecting communication. Note, however, that dependence does not imply communication: agents that share the same underlying model, system prompt or task can exhibit correlated outputs without exchanging any information, so the total correlation of a non-communicating baseline must be measured and subtracted. Moreover, estimating the joint entropy is considerably harder than estimating the marginal entropies, because the joint outcome space grows exponentially with the number of agents and the vocabulary size, and plug-in estimators are biased with limited samples (Paninski, 2003). For completeness, in this project we compute both the sum of the individual entropies (the independence assumption) and an estimate of the TC term. We restrict our operational analysis to a two-agent system ($n = 2$), where the total correlation $TC(X_1; X_2)$ reduces directly to the pairwise mutual information $I(X_1; X_2)$ (Cover & Thomas, 2006).

Specifically, for $n = 2$ agents, Equation (4) becomes

$$H(X_1, X_2) = H(X_1) + H(X_2) - I(X_1; X_2), \quad (5)$$

where each individual entropy $H(X_i) = -\sum_{v \in V} p_i(v) \log_2 p_i(v)$ quantifies the local decision uncertainty of agent i over its full next-token distribution, obtained with the softmax of Equation (2) applied to the raw logits ($T = 1$) and averaged over the tokens of its response. Top- k renormalised distributions are used only as a fallback when full logits are not available, and yield a lower bound that is not comparable with full-vocabulary values. The interaction term $I(X_1; X_2)$ is the mutual information that couples the two agents,

$$I(X_1; X_2) = \sum_{x_1 \in \mathcal{X}_1} \sum_{x_2 \in \mathcal{X}_2} p(x_1, x_2) \log_2 \left(\frac{p(x_1, x_2)}{p(x_1)p(x_2)} \right) \quad (6)$$

which is the Kullback–Leibler divergence D_{KL} between the joint distribution $p(x_1, x_2)$ and the product of the marginals $p(x_1)p(x_2)$ (Cover & Thomas, 2006).

Under this framework, if the two agents operate independently, $p(x_1, x_2) = p(x_1)p(x_2)$, so $I(X_1; X_2) = 0$ and the system entropy is purely additive, $H(X_1, X_2) = H(X_1) + H(X_2)$. Conversely, when the

agents interact, align or exchange information, $p(x_1, x_2) \neq p(x_1)p(x_2)$ and the mutual information becomes positive, $I(X_1; X_2) > 0$. This term subtracts the uncertainty shared by the pair, so the joint entropy $H(X_1, X_2)$ falls below the sum of the individual entropies, reflecting the emerging order and statistical coupling of the two-agent collective (Cover & Thomas, 2006; Watanabe, 1960).

4. Results

Present your main findings with appropriate evidence. Use figures and tables where appropriate (we strongly encourage at least one figure, see tips below). Distinguish between observations and interpretations.

Argue why your claims are robust. E.g., if your approach “performs better” than alternatives:

- (a) *Do you have enough data? Is the difference statistically significant?*
- (b) *Is it robust? Or do small changes to your setup cause substantial changes to your results?*

Tips for figures and tables:

- *Number all figures (Figure 1, Figure 2...) and tables (Table 1, Table 2...)*
- *Include descriptive captions that can be understood without the main text*
- *Place figures/tables near where they’re first referenced*
- ***Ensure text in figures is legible!***

5. Discussion and Limitations

Discuss the broader implications for AI safety.

What do your results mean? What trends do you notice and what might they indicate?

Limitations

The proposed method uses the predictive entropy of the agents, and the coupling between them captured by the total correlation, as a proxy for changes in behaviour, with a particular focus on the emergence of communication within a MAS. Its purpose is to provide a channel-agnostic signal, since the improvised message boards used by the agents in the OpenAI–Hugging Face incident were hosted on resources never intended for communication. However, the same breadth that makes the measure channel-agnostic also makes it sensitive to other sources of variation, such as the normal dynamics of agents while solving a task, response length, prompt formatting and temperature. In addition, statistical dependence does not imply communication, since agents sharing a model, prompt or task may be correlated without exchanging any information; joint entropy is hard to estimate with limited samples; and many APIs expose only top- k log-probabilities, so the entropy can only be approximated. The method also requires access to the models’ output distributions, which restricts it to the operator hosting the agents, and our experimental evidence is preliminary, based on a single simple benchmark.

Future Work

Further research is therefore needed to detect specific behaviours from the entropy of the system more precisely. A particularly promising direction is the design and training of a machine learning model on entropy trajectories that distinguishes communication from other behavioural changes. Beyond the output distribution, a complementary direction is to exploit the internal activations of the models: lightweight probes trained on the hidden states of intermediate layers have already been deployed to detect harmful inputs in production at a small fraction of the cost of LLM-based classifiers (Kramár et al., 2026), and similar activation-based monitors could be combined with, or replace, entropy signals to detect communication and other unintended behaviours in a MAS more directly, although they would likewise require white-box access to the models. It is also necessary to study more broadly how tools, new information and other perturbations affect the entropy of a MAS, to establish non-communicating baselines across architectures and models, and to validate the approach in more realistic settings, with the aim of building a more robust detector of unintended behaviours and capabilities in general.

6. Conclusion

Briefly summarize your main findings and their implications (1–2 paragraphs).

Code and Data

Include links if applicable. If your project doesn't involve code (e.g., policy analysis) or if there are info-hazard considerations, note that here.

- **Code repository:** *[Link to GitHub/GitLab if applicable]*
- **Data/Datasets:** *[Link if applicable]*
- **Other artifacts** (optional): *[Demo link, video walkthrough, Hugging Face Space, etc.]*

Author Contributions (optional)

[e.g., “J.C.M. led the project and designed experiments. A.S.P. implemented the code. All authors contributed to writing and reviewed the final manuscript.”]

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Appendix (optional)

Supplementary material such as additional figures, detailed methodology, prompts used, extended results, etc.

LLM Usage Statement

If you used LLM assistance in developing your project or writing this report, briefly note how. Ensure all claims and results have been verified.

NOTE: We strongly encourage that the final version of the submission is primarily written by your team.

[e.g., “We used Claude to brainstorm approaches and help draft sections. All results and claims were independently verified.”]

We used some AI assistance in the development of the bulk code for the project. Without letting it define the infrastructure, nor deciding the direction of the work. We’ve also used as an style corrector for the document. All claims were independently verified and contrasted.